PRIVATE AVIATION OPERATIONS PLATFORM

FlightDesk AI

A working prototype: customer intake, team workflow, and an AI assistant with built-in safety and measured accuracy.

This document explains the whole project in plain English, in one read. It's written so you can walk someone through it -- what it does, how the AI is kept safe, and why the decisions were made -- without needing to look at the code.

> Independent portfolio prototype, not an official Flygreen product and not affiliated with Flygreen or any aircraft operator. All customers, quotes and bookings shown in the demo are fictional. No real email is sent and no real bookings are made.

1. The one-sentence pitch

> A private-jet charter business gets requests, prepares quotes, follows up with customers, and tracks bookings -- this app does all of that, plus an AI assistant that can answer questions and draft work for a human to approve, never acting on its own.

2. Who uses it (three roles)

- **Customer:** Submits a flight request on a public form. No login needed. Can check their request status later.
- **Team member:** Signs in to an Inbox showing new requests, quotes to send, and follow-ups due. Can draft quotes and emails, request quotes from operators, but can't see money or confirm bookings.
- **Business owner:** Everything the team has, plus a dashboard (money, conversion, trends), reports, AI evaluation results, user management, and the power to confirm bookings or delete data.

3. The core workflow, step by step

1. A customer fills in a flight request form (route, dates, passengers) on the public site.
2. It lands in the team's Inbox as a new inquiry, with AI-extracted trip details to double-check.
3. The team member requests quotes from aircraft operators (WhatsApp or email, pre-written message).
4. Operator replies come back; quotes get logged, compared side by side, and the best one is picked.

5. A follow-up email or quote PDF is drafted (by AI or template) and sent to the customer for review.
6. Once the customer agrees, the business owner confirms the booking -- this step needs a human, always.
7. The owner's dashboard shows live numbers: new inquiries, conversion rate, revenue trend, follow-ups due.

How the AI is used (and kept safe)

The AI (OpenAI's models) is used in several places, but it never acts on its own. Every AI-written action -- an email, a status change, a booking -- is shown to a human as a proposal that must be approved before anything is saved or sent.

Where AI is used

- **Reading messages:** Turns a pasted customer email or an operator's PDF quote into structured data (route, dates, price), with a confidence check before it's accepted.
- **Assistant chat:** A business owner or team member can ask questions in plain English ('what needs my attention today?') and the AI answers using live data from the app, not guesses.
- **Drafting:** Writes follow-up emails and quote summaries in a chosen tone, which a human edits and approves before anything is sent.
- **Voice:** Speech-to-text lets you talk to the assistant instead of typing.

The safety rules (this is the important part)

- **Approval required:** Any AI-suggested change (status update, booking, email) is a 'proposal' that sits waiting until a human clicks Approve. The AI cannot write to the database directly.
- **Role-based access:** Team members cannot see financial figures or confirm bookings, even by asking the AI -- those tools are switched off for their role at the system level, not just hidden in the UI.
- **Grounded answers:** The assistant is instructed to answer only from real data pulled from the database, and is evaluated specifically on whether it invents facts.
- **No destructive actions by AI:** Deleting data, confirming bookings, and similar high-risk actions are either blocked for the AI entirely or require explicit human confirmation with a typed phrase.

How we know it actually works: evaluation

Rather than just trusting the AI, this project includes an evaluation system that runs real test cases against the assistant and scores the answers automatically -- checking for correctness, safety, and whether it stays within its role permissions. Results (pass rate, failure categories, cost per run) are tracked over time so changes can be measured, not just assumed to be better.

Features by area

Customer-facing

- **Public request form:** No account needed. Clean, mobile-friendly form to submit a trip request.
- **Status check:** Customers can look up their request status without logging in.

Team workspace

- **Inbox:** A prioritized to-do list: new inquiries, quotes awaiting review, follow-ups due.
- **Inquiries & quotes:** Structured records for every request, with side-by-side quote comparison.
- **Operators:** A directory of aircraft operators; one click drafts a quote-request message ready for WhatsApp or email, and tracks who has replied.
- **Follow-ups:** Templated or AI-drafted emails, queued for approval, with a due-date tracker.
- **Documents:** Generates branded PDF quotes for customers and can read PDF quotes sent by operators.
- **Customers:** A simple CRM view of every customer and their inquiry history.

Business owner tools

- **Dashboard:** Live KPIs: new inquiries, conversion rate, revenue trend, outstanding follow-ups.
- **Reports:** Deeper breakdowns for performance review.
- **AI Evaluation Lab:** Run and review automated accuracy/safety tests against the assistant.
- **Settings:** Manage users, roles, and system configuration.

Under the hood (if asked)

- **Stack:** Python, Streamlit for the interface, SQL database, OpenAI API for AI features.
- **Structure:** Cleanly separated: 'services' hold business logic, 'ui' holds the screens -- so logic isn't tangled with how it's displayed.
- **Data model:** Inquiries, Quotes, Customers, Operators, Bookings, Follow-ups, Users -- all linked relationally so the dashboard and AI assistant draw from one consistent source of truth.

How to explain it in 30 seconds

> "It's a working prototype of a charter-aviation operations tool. A customer submits a flight request, the team gets quotes from operators and sends follow-ups, and the owner tracks everything on a dashboard. An AI assistant helps throughout -- reading messages, drafting emails, answering questions -- but it can never take action by itself: every AI suggestion needs a human to approve it first. I also built an evaluation system to actually measure how accurate and safe the AI is, instead of just assuming it works."